

Project R&D

Goals

React application that:

- Uploads PDFs
- Reads text from scanned PDFs/images using OCR or google vision api
- Supports multiple languages
- Generates accurate summaries in any language
- Includes zoom in / zoom out
- Supports page navigation
- Allows text selection and copy
- Works with scanned and digital PDFs
- Has clean UI and responsive design

Flow

Upload pdf — render — extract text – optical character recognition if needed — convert text into useful points — send it to AI — generate Summary — translate — Display

1.Pdf rendering

Library :

react-pdf

2. Optical Character Recognition (Image to text)

the technology used to convert images of typed, handwritten, or printed text—such as scanned documents, photos, or PDFs—into editable, searchable, and machine-encoded text

Library:

[Tesseract.js](#)

Supports:

- English
- Hindi
- Japanese
- Chinese
- Arabic
- Many more

Features:

- Client-side OCR
- Works in React
- Multi-language support
- Decent accuracy

Google Vision API

Google Vision API is an AI-powered image analysis service developed by Google that can extract and understand information from images and PDFs

Google Vision API will:

- Detect text
- Recognize handwriting
- Extract structured content
- Return clean text data

3.AI Summary System

API:

OpenAI API / Gemini API or any with free tokens

Architecture

Frontend:

1. PDF upload
2. Render PDF
3. Zoom controls
4. OCR processing UI
5. Language selection
6. Summary display
7. Loading states

Backend:

1. AI API calls
2. Chunk processing
3. Summary generation
4. Translation

Stack

Frontend

1. React
2. Vite
3. Tailwind
4. react-pdf
5. Tesseract.js

Backend

1. Node.js
2. Express
3. OpenAI API

